

Foundations of MAAT Structural Supports

Why H , B , S , V , Log-Support Selection, and Emergent Robustness

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Abstract

The MAAT structural-selection series uses four primitive diagnostic support coordinates (H, B, S, V) , a bounded support map $\Gamma_a = 1/(1 + d_a)$, a logarithmic selection functional, and an emergent robustness closure

$$R_{\text{rob}} = \min\left((HBV)^{1/3}, (HBSV)^{1/4}\right).$$

Previous papers derived the exponential selection measure from maximum-entropy reasoning once structural defects are specified, and later papers developed local defect-field geometry and multi-scale projections. What remained open was the more basic foundations question: why these supports, why this support map, why the logarithmic form, and why robustness should be emergent rather than a fifth primitive diagnostic coordinate.

This paper gives a deliberately conservative answer. It does not claim a microscopic derivation of MAAT. Instead, it formulates an operational axiomatic basis: a structural-selection layer must test admissibility consistency, constraint balance, controlled activity, and coherent coupling. These four requirements motivate H, B, S, V as primitive diagnostic support coordinates, not as new physical fields or independent dynamical degrees of freedom. The bounded support map follows from monotonicity, normalization, and comparability requirements. The logarithmic functional follows when independent sector supports multiply, or equivalently from the maximum-entropy exponential family applied to defect costs. The robustness closure is then a bottleneck condition: passive adherence $(HBV)^{1/3}$ must survive activity-sensitive stress $(HBSV)^{1/4}$.

The result is not a proof that MAAT is fundamental. It is a foundations specification: a clear statement of what is assumed, what is derived conditionally, and how the assumptions can fail. The foundation is therefore not a claim of final ontology, but a predeclared coordinate system for structural tests.

1 Purpose

The purpose of this paper is to address an open foundational point in the MAAT series. Many earlier papers used the structural supports

$$H, \quad B, \quad S, \quad V,$$

the support map

$$\Gamma_a = \frac{1}{1 + d_a},$$

and the selection functional

$$F_{\text{MAAT}}[X] = - \sum_a \lambda_a \log(\epsilon + \Gamma_a[X]).$$

Paper 48 showed that exponential structural measures follow from a maximum-entropy argument once a set of structural defects $d_a[X]$ has been chosen [7]. Paper 56 formalized local defect fields and multi-scale projections once those sectors are operationalized [10].

However, those results do not by themselves answer the deeper question:

Why should the primitive diagnostic structural language be built from consistency, balance, activity, and connectivity, with robustness as an emergent closure?

This paper answers that question at the level of operational foundations, not microscopic derivation. The claim is modest:

If one wants a domain-general diagnostic layer for structural selection, then H, B, S, V form a minimal four-axis basis for admissibility, constraint closure, nontrivial dynamical richness, and coherent coupling.

Diagnostic status. Throughout this paper, H, B, S, V should be read as diagnostic support coordinates defined over an already specified dynamical, logical, or empirical system. They are not introduced as new physical fields, new particles, new fundamental forces, or independent degrees of freedom. Likewise, R_{resp} and R_{rob} are derived closure diagnostics over those supports, not additional primitive variables.

2 What Is Not Proven Here

The paper deliberately separates three levels:

1. **Conditional mathematical form.** Given defect constraints, maximum entropy gives an exponential measure. This part is mathematical.
2. **Operational diagnostic support basis.** The coordinates H, B, S, V are motivated by necessary diagnostic roles. This part is axiomatic and falsifiable.
3. **Microscopic origin.** A derivation of the sectors from an underlying action, symmetry principle, renormalization flow, entropy-production law, or quantum information principle is not yet available.

Thus this paper does not claim that MAAT is a completed fundamental theory. It makes the assumptions explicit so they can be tested, replaced, or refined.

3 Structural-Selection Requirements

Let $\mathcal{X}_{\text{adm}}$ be a set of dynamically or logically admissible configurations. The word “admissible” means that configurations have passed the basic domain rules: field configurations satisfy boundary conditions, SAT instances are well-formed, cosmological trajectories solve the chosen background equations, and candidate unified models obey their declared input assumptions.

A structural-selection layer ranks configurations inside $\mathcal{X}_{\text{adm}}$. To be useful across domains, it must answer four different questions.

Sector	Question	Failure if omitted
H	Is the candidate internally consistent with its equations or local rules?	Inconsistent or equation-violating candidates can be ranked highly.
B	Are competing constraints, resources, signs, charges, or conservation channels balanced?	One constraint can be satisfied by hiding violation in another channel.
S	Is there controlled activity, richness, or nontrivial dynamical capacity?	The ranking collapses toward frozen, trivial, or inactive solutions.
V	Are the relevant degrees of freedom coherently coupled or connected?	Disconnected artifacts can look locally good while failing globally.

These four questions motivate the primitive diagnostic support basis:

$$\boxed{\mathcal{A}_{\text{prim}} = \{H, B, S, V\}}. \quad (1)$$

The basis is not claimed to be unique in a mathematical classification sense. It is claimed to be minimal in the following operational sense: removing any one sector leaves a known class of structural failure undetected.

Operational minimality claim. For each diagnostic coordinate

$$a \in \{H, B, S, V\},$$

there exists a class of admissible candidates $\mathcal{X}_{\neg a}$ such that the remaining supports can be high while the omitted failure mode is not detected. Thus minimality means diagnostic non-redundancy, not mathematical uniqueness.

Omitted coordinate	Failure class that can remain hidden
$\neg H$	A candidate can be balanced, active, and connected while violating the equations, rules, or target relations that define admissibility.
$\neg B$	A candidate can satisfy one visible constraint while exporting stress into a conservation, symmetry, threshold, charge, or side-condition channel.
$\neg S$	A candidate can be consistent, balanced, and connected while being frozen, trivial, dynamically inactive, or unable to generate useful variation.
$\neg V$	A candidate can be locally consistent in isolated components while failing as an integrated whole because the components are weakly coupled or disconnected.

4 Why These Four Sectors?

4.1 H: Consistency

The H sector measures whether a candidate satisfies the local rules that make it admissible as a configuration of the system. In field theory this can be distance from equations of motion. In

cosmology it can be residual background consistency. In SAT it can be local clause or propagation consistency. In machine-learning hardness tests it can be local label or manifold consistency.

Without H , structural selection can rank configurations that are rich, connected, and balanced but simply wrong. Thus H is the admissibility anchor.

4.2 B: Balance

The B sector measures whether multiple constraints are jointly respected. Typical examples include conservation laws, resource balances, symmetry constraints, charge or tadpole closure, class balance, or competing objective channels.

Without B , a candidate can appear coherent by over-satisfying one channel while exporting stress into another. Thus B is the constraint-closure anchor.

4.3 S: Controlled Activity

The S sector measures useful dynamical richness. It is not raw activity. In many systems both too little and too much activity are structurally bad: too little gives frozen triviality, while too much gives instability, noise, or chaos. This is why later MAAT papers often use an effective activity window

$$S_{\text{eff}}(A) = \exp \left[-\frac{1}{2} \left(\frac{A - A_{\star}}{\sigma_A} \right)^2 \right],$$

rather than rewarding activity without bound.

Without S , structural selection tends to prefer static coherence over productive dynamics. Thus S is the nontriviality anchor.

4.4 V: Connectivity

The V sector measures coherent coupling between relevant degrees of freedom. Depending on the domain, this can mean graph connectivity, field coupling, correlation, information transfer, growth-lensing coherence, or shared low-dimensional residual modes.

Without V , a configuration can look locally consistent in each component while failing to function as a connected whole. Thus V is the integration anchor.

5 Minimality Does Not Mean Uniqueness

The preceding argument is a minimality argument, not a uniqueness theorem. It says that each of the four diagnostic coordinates performs a distinct role whose removal creates a recognizable blind spot. It does not say that no other coordinates can be useful.

Several alternative diagnostic languages are plausible:

Alternative axis	Why it is natural	MAAT status
Entropy production	Irreversibility and dissipation are central in non-equilibrium systems.	Usually enters as an activity or balance diagnostic; it may become a primitive diagnostic coordinate in thermodynamic domains.
Symmetry preservation	Symmetries organize physical theories and constrain allowed states.	Usually enters through H and B : equations and constraints must respect declared symmetries.
Scale invariance	Many systems are governed by renormalization or multi-scale structure.	Treated as a projection/coarse-graining property in the v1.4 defect-field extension, not as a universal primitive diagnostic coordinate.
Information preservation	Coherent information flow is essential in quantum, ML, and network systems.	Often appears through V and H ; it can define a domain-specific operationalization of connectivity.
Stability alone	Persistence is the observable outcome one often cares about.	Treated as emergent R_{rob} , because using it as a primitive risks double counting the closure channels that generate stability.

This gives the correct status of the H, B, S, V diagnostic basis:

It is the canonical MAAT basis because it separates four recurring failure modes, not because alternatives are impossible.

An alternative diagnostic basis should replace or extend H, B, S, V if it satisfies all of the following conditions:

1. it gives operational definitions before seeing benchmark labels;
2. it is less redundant under covariance and ablation tests;
3. it improves prediction or ranking against domain-native baselines;
4. it survives shuffled-defect and leave-family-out controls;
5. it transfers at least as well across declared domains without retuning.

Thus the foundations are not closed by definition. They define a default coordinate system and a replacement rule.

6 Why Robustness Is Not Primitive

Early versions treated robustness or respect as a fifth primary diagnostic coordinate. The v1.2.1 closure changed this:

H, B, S, V are primitive diagnostic coordinates, R is emergent.

The reason is double counting. Robustness is not independent of consistency, balance, activity, and connectivity. A system is robust because it maintains consistency, balance, and coupling under activity and perturbation. If R is introduced as an additional unconstrained primitive diagnostic score, it can repackage information already present in H, B, S, V .

The passive response closure is

$$\boxed{R_{\text{resp}} = (HBV)^{1/3}}. \quad (2)$$

It measures structural adherence under consistency, balance, and connectivity, before activity stress is included.

The activity-sensitive robustness closure is

$$\boxed{R_{\text{rob}} = \min\left(R_{\text{resp}}, (HBSV)^{1/4}\right)}. \quad (3)$$

The minimum is important. Robustness is a bottleneck property: it cannot exceed the weakest relevant closure channel. The first term enforces passive adherence; the second term asks whether that adherence survives controlled activity.

Future extensions may define an active-significance diagnostic

$$R_{\text{sig}} = R_{\text{resp}}^{1-\alpha} S_{\text{eff}}^{\alpha},$$

but this is not a new primitive coordinate. It is a derived observable over the primitive diagnostic support language.

Equation (3) should therefore be read as a canonical robustness closure, not as a mathematical uniqueness theorem. Other closures, such as soft-min functions, weighted geometric means, or domain-specific risk envelopes, are legitimate competitors. They become preferable if they improve out-of-sample robustness prediction without reintroducing a freely fitted fifth primary diagnostic coordinate.

7 Why Bounded Supports?

Let each primitive diagnostic coordinate have a non-negative defect

$$d_a[X] \geq 0, \quad a \in \{H, B, S, V\}.$$

The support $\Gamma_a[X]$ should satisfy:

1. **Normalization:** zero defect gives maximal support;
2. **Monotonicity:** larger defect gives lower support;
3. **Comparability:** supports from different domains live on a common bounded scale;
4. **No singular reward:** perfect support should be finite;
5. **Small-defect linearity:** small defects should behave like ordinary additive costs.

Defect normalisation protocol. The map $\Gamma_a = 1/(1 + d_a)$ is meaningful only after each defect has been made dimensionless and comparable. Each benchmark should therefore declare

$$d_a[X] = \frac{d_a^{\text{raw}}[X]}{\sigma_a}, \quad (4)$$

or an equivalent dimensionless normalization, before final evaluation. The scale σ_a may be chosen from measurement uncertainty, physical tolerance, a reference-ensemble scale, a validation-set

quantile, or a domain-native error scale. It must not be tuned after inspecting the final benchmark labels.

The canonical MAAT choice is

$$\boxed{\Gamma_a[X] = \frac{1}{1 + d_a[X]}, \quad 0 < \Gamma_a \leq 1.} \quad (5)$$

This map is not the only possible monotone map. Alternatives such as $\exp(-d_a)$ or logistic supports can be tested. The rational map is used because it is minimal, bounded, scale-stable, and has the expansion

$$-\log \Gamma_a = \log(1 + d_a) = d_a + O(d_a^2).$$

Thus the support representation agrees locally with a linear defect cost while remaining globally bounded.

The rational map is therefore a canonical gauge choice for support geometry, not a claimed necessity. If another monotone map gives better calibrated probabilities, stronger cross-domain transfer, or lower null-model leakage under the same declared defects, then the support geometry should be updated. The invariant part of the framework is not the exact rational form; it is the requirement that defects become comparable bounded viability supports.

8 Why the Log-Support Functional?

If the supports of different structural sectors contribute multiplicatively to viability, then

$$P[X] \propto \prod_a (\epsilon + \Gamma_a[X])^{\lambda_a}.$$

Taking the negative logarithm gives

$$\boxed{F_{\text{MAAT}}[X] = - \sum_{a \in \{H, B, S, V\}} \lambda_a \log(\epsilon + \Gamma_a[X]).} \quad (6)$$

This is the support-form version of the maximum-entropy exponential family. Indeed,

$$P[X] \propto e^{-F_{\text{MAAT}}[X]}.$$

For small defects and small ϵ ,

$$-\lambda_a \log(\epsilon + \Gamma_a[X]) \simeq \lambda_a d_a[X]$$

up to constants and higher-order corrections. Thus the log-support form is not an arbitrary aesthetic choice. It is the natural additive cost associated with multiplicative viability supports, and it is locally equivalent to the linear MaxEnt defect functional.

Regularisation convention. The parameter ϵ is a numerical regularizer, not a fit parameter. It prevents singular logarithms when a support approaches zero. Main benchmark claims should use a fixed declared value of ϵ and report whether rankings remain stable under sensitivity checks such as

$$\epsilon \in \{10^{-12}, 10^{-9}, 10^{-6}\}.$$

If conclusions depend sensitively on ϵ , the result should be reported as regularization-dependent.

9 Why Response-Closed Weights?

The coefficients λ_a are not meant to be decorative fit knobs. In the MaxEnt derivation they are Lagrange multipliers conjugate to the chosen structural constraints [1, 2]. Operationally, later MAAT papers estimate them through defect covariance:

$$\lambda \approx \left(C + \eta \frac{\text{tr}(C)}{A} \mathbb{I} \right)^{-1} (\langle d \rangle_{\mu_0} - \langle d \rangle_*), \quad C = \text{Cov}_{\mu_0}(d). \quad (7)$$

The response-closure idea is simple: weights should reflect how strongly a sector responds when the ensemble is moved from a reference state toward a target low-defect or safe subensemble.

For canonical structural ranking, weights are constrained to be non-negative after orientation of the defects. Signed response weights can be useful diagnostics, but if a defect called an “obstruction” receives a stable negative selection pressure, this is evidence that the defect orientation, reference ensemble, or target set is misdefined.

10 Field-Functional Extension

The scalar functional in Eq. (6) is the macro-projection of a richer defect-field functional. For local domains one should write

$$d_a = d_a(x), \quad \Gamma_a = \Gamma_a(x),$$

and define multi-scale costs such as

$$F_{\text{multi}} = F_{\text{mean}} + F_{\text{tail}} + F_{\text{cluster}} + F_{\text{scale}}. \quad (8)$$

This is not a rejection of the scalar MAAT form. It clarifies its domain of validity. Smooth, mean-field, or background cosmological problems can often be described by scalar supports. SAT hardness, spin-glass frustration, local quantum pointer conflicts, and graph-local search dynamics may require tails, clusters, and scale projections.

Thus the scalar F_{MAAT} is the low-frequency structural score, while the defect-field formulation is the higher-resolution extension.

11 Generator-Aware Supports

For evolving systems, a static support coordinate may miss the object that actually defines persistence. The relevant question is not only what a configuration looks like, but what the domain generator preserves, damps, transports, or destabilizes.

Let $\mathcal{G}$ denote the generator of the domain: a time-evolution operator, RG flow, Markov transition rule, Lindblad generator, solver transition, or search policy. A generator-aware support coordinate has the schematic form

$$H_{\mathcal{G}}, B_{\mathcal{G}}, S_{\mathcal{G}}, V_{\mathcal{G}} = \mathcal{D}_{H,B,S,V} (X, \mathcal{G}X, \mathcal{G}^2X, \dots). \quad (9)$$

This is the foundations-level version of generator stationarity used in later dynamical benchmarks. Static H, B, S, V are macro supports. Generator-aware supports are the dynamical refinement: they test whether the relevant structural property is stable under the process that actually acts on the system.

12 Falsifiability

The diagnostic support basis is useful only if it survives comparison against simpler alternatives. The following tests should be treated as mandatory whenever data are available:

1. **Energy-only or action-only baselines** in physical systems;
2. **Entropy-only, stability-only, or complexity-only baselines** where appropriate;
3. **Domain-native baselines** such as NFW/MOND diagnostics in galaxy rotation curves, CDCL heuristics in SAT, or decoherence-theory predictors in quantum pointer tests;
4. **Shuffled-defect nulls** that preserve marginals but destroy sector coherence;
5. **Leave-family-out or cross-domain transfer tests** without retuning.

If H, B, S, V and the induced R_{rob} do not outperform these baselines under predeclared metrics, then the correct conclusion is not that MAAT is hidden. The correct conclusion is that the current diagnostic support language is non-informative for that benchmark.

13 Empirical Motivation, Not Proof

The existing benchmark record is useful for foundations only if it is read correctly. It does not prove the primitive diagnostic support basis. It does, however, identify the kinds of successes, failures, and resolution effects that any foundation of MAAT must explain.

Benchmark layer	Representative result	Foundational lesson
Field validation	In the minimal ϕ^4 benchmark, structural ranking outperforms energy ranking at the tested percentile and separates kink-like structure from distortions [6].	The support language is not automatically reducible to energy, at least in controlled field benchmarks.
Universality audit	The benchmark-readiness index is low (0.3816), with complete competitor tests available only in part of the domain set [8].	The foundations must remain falsifiable; recurrence across domains is not yet a universality proof.
SAT hardness	Graph-local MAAT+density improves over density-only in Paper 50, but simple graph baselines remain competitive [9].	Global scalar compression is insufficient for local frustration geometry; defect-field resolution matters.
Active SAT search	Mode-A branching has mixed regret against VSIDS, while structure-gating reduces tail damage in the next benchmark [11, 12].	Supports can act as policy coordinates, but only when anchored by instance-level structure diagnostics.
Quantum pointer tests	Two-qubit pointer robustness is predicted better by correlated stationarity and field features than by simpler scalar population balance [13].	For evolving systems, generator stationarity can be more fundamental than static balance alone.
SPARC rotation curves	The SPARC-II pilot finds a shuffled-radius significant NFW-like radial signal, while the RAR proxy is not significant in the same test [14].	External data can carry structural signal, but the signal is model- and diagnostic-dependent rather than automatically universal.

These results motivate the present foundations in a narrow way. They support three conservative lessons:

1. structural supports can add information beyond some simple scalar baselines;
2. scalar MAAT is not sufficient in all domains;
3. the diagnostic support basis must be evaluated at the right structural resolution and against strong domain-native controls.

Thus the empirical record strengthens the need for an explicit support foundation, but it does not replace that foundation with evidence by accumulation.

14 Relation to Earlier Papers

Text	Role
Core Formalism Primer	Defines the common notation and master architecture.
Paper 36	Establishes the v1.2.1 closure: R becomes emergent.
Paper 48	Derives the exponential/log-support form from MaxEnt once defects are specified.
Paper 56	Extends scalar scores to local defect-field geometry and multi-scale projections.
This paper	Gives the operational foundations for why the primitive diagnostic support basis is H, B, S, V and why R_{rob} is a closure diagnostic.

15 Limitations

The foundations given here are intentionally limited.

- They do not derive H, B, S, V from a microscopic action.
- They do not prove that the rational support map is unique.
- They do not prove that R_{rob} is the only possible robustness closure.
- They do not replace empirical validation.
- They do not imply that every domain admits a useful scalar MAAT score.

The main value of the paper is therefore not finality. Its value is that the assumptions are now explicit enough to be attacked.

16 Conclusion

The MAAT support language can be justified at three different strengths. At the strongest level, not yet achieved, one would derive the sectors from microscopic physics or a deeper information principle. At the weakest level, one merely postulates useful scores. This paper occupies the middle level:

H, B, S, V are motivated as a minimal operational diagnostic basis for structural selection, the bounded support map makes defects comparable, the logarithmic functional follows from multiplicative viability and MaxEnt, and robustness is an emergent bottleneck closure diagnostic rather than a fifth primitive degree of freedom.

The resulting framework should be treated as a canonical default, not as a closed ontology. If another declared support basis, support map, or robustness closure outperforms it under the falsification rules above, MAAT should absorb that failure as a refinement of its coordinate language.

The foundation is therefore not a claim of final ontology, but a predeclared coordinate system for structural tests.

This does not prove MAAT. It makes MAAT testable at its own foundations.

Reproducibility and Repository

This paper is conceptual and does not introduce a new numerical benchmark. The LaTeX source, compiled PDF, and companion code for the broader structural-selection series are maintained in:

<https://github.com/Chris4081/structural-selection-principle>

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